

RAJAT BISHT

SOFTWARE | DATASCIENCE | CREATIVE TECHNOLOGY

+91-8126432393 · rajatbisht913@gmail.com · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

PROJECTS

AI-Driven Fraud Detection System

github.com/rb937/fraud-detection-system

Nov 2025 – Jan 2026

Python · Scikit-Learn · Pandas · Streamlit

- Engineered a real-time fraud detection pipeline using Random Forest, achieving 92% Recall on a heavily imbalanced dataset via SMOTE oversampling and cost-sensitive learning.
- Developed geospatial risk features using Haversine Distance to flag location anomalies (transactions >500 km from home) and deployed findings as an interactive Streamlit dashboard.

ChromaEngine – Image Processing Engine

github.com/rb937/ChromaEngine

Dec 2025 – Feb 2026

Java · JavaFX · OOP

- Built a non-destructive image processing engine using direct-memory bitwise operations on ARGB channels, eliminating BufferedImage overhead for real-time performance.
- Implemented custom 3x3 convolution matrices (sharpen, blur, edge detection) to automate day/night cycle and lighting variant generation for 2D pixel-art asset pipelines.

Terminal Snake Engine

github.com/rb937/terminal-snake-C

Feb 2025

C · Windows API · Systems Programming

- Implemented flicker-free rendering using Windows Console API cursor repositioning, eliminating system("cls") and achieving smooth high-framerate terminal output.
- Engineered velocity normalisation to correct terminal font aspect-ratio distortion and a dynamic difficulty curve with a terminal-velocity floor to maintain playability at high scores.

Wireless ESP32 Smart Car

Nov 2023

Embedded C · ESP32 · IoT · Hardware

- First Prize — Youthopia, DIT University (Robo Soccer). Engineered a wireless-controlled bot with superior navigation and motion stability, outperforming competing teams.
- Programmed ESP32 firmware in Embedded C to integrate motor drivers and GPIO controls for real-time responsiveness under competition conditions.

SKILLS

Languages	Python, Java, C, R, SQL
ML / Data	Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn
Tools	Streamlit, Git, MS Excel, Power BI
Creative	Adobe Photoshop, Premiere Pro, Blender 3D
Hardware	Arduino, ESP32
Game Engines	Unity, Godot

EDUCATION

B.Tech — Computer Science Engineering (Data Science)
DIT University, Dehradun · CGPA: 8.03 / 10

2023 – 2027

CERTIFICATIONS

Programming Fundamentals – Duke University
Database Management Essentials – University of Colorado
Data Fundamentals – IBM

EXTRACURRICULAR ACTIVITIES

- Produced marketing reels and event highlight videos for the IEEE club at DIT University as part of the core video editing team.
- Organised multiple technical and cultural events as an active IEEE club member.